

Factors related to the prevalence of pathogenic Yersinia enterocolitica on pig farms.



A survey of 788 pigs from 120 farms was conducted to determine the within-farm prevalence of pathogenic *Yersinia enterocolitica* and a questionnaire of management conditions was mailed to the farms afterwards. A univariate statistical analysis with carriage and shedding as outcomes was conducted with random-effects logistic regression with farm as a clustering factor. Variables with a P value <0.15 were included into the respective multivariate random-effects logistic regression model. The use of municipal water was discovered to be a protective factor against carriage and faecal shedding of the pathogen. Organic production and buying feed from a certain feed manufacturer were also protective against total carriage. Tonsillar carriage, a different feed manufacturer, fasting pigs before transport to the slaughterhouse, higher-level farm health classification, and snout contacts between pigs were risk factors for faecal shedding. We concluded that differences in management can explain different prevalences of *Y. enterocolitica* between farms.